Your Cancer Summary Report

Jane Doe

Stage IIIC High-Grade Serous Ovarian Cancer (HGSOC)

HGSOC Precision Oncology Research Center

Report Date: March 10, 2026

⚠️ **DRAFT - FOR CLINICIAN REVIEW**

This report has not yet been reviewed by your healthcare team. Please wait for your doctor to discuss these results with you.

At a Glance

Your Diagnosis

Ovarian Cancer - High-Grade Serous Carcinoma (HGSOC)

Stage: Stage IIIC | **Grade:** High Grade

You have been diagnosed with high-grade serous ovarian cancer, an advanced cancer that has spread within the abdomen. Your medical team has run detailed molecular and spatial tests to identify the best treatment options for your specific cancer.

✓ Positive Factors

- HRD score of 72 (above the 42 threshold) means PARP inhibitor maintenance therapy is recommended — these drugs can significantly delay recurrence
- Strong antigen presentation score (0.85) means your immune system may respond well to immunotherapy
- Five clinical trials currently recruiting that match your molecular profile

⚠ Challenges to Consider

- CCNE1 amplification is a marker of potential platinum resistance — your care team will monitor response to first-line chemotherapy closely
- CD8+ immune cells are present but partially excluded from the tumor core — additional therapies may be needed to help them penetrate the tumor

What Your Tests Found

Genetic testing of your tumor revealed the following important findings:

TP53

Pathogenic

Copy number loss (log2 = -0.65)

TP53 is the main 'guardian of the genome' gene. Loss of TP53 is found in nearly all HGSOC tumors and confirms the high-grade serous subtype. It also makes your tumor a candidate for neoantigen vaccines targeting TP53 mutations.

Treatment Option: Investigational neoantigen vaccine (RMPEAAPPV peptide, IC50 = 7.8 nM, strong HLA-A*02:01 binder)

BRCA2

Likely Pathogenic

Copy number loss (log2 = -0.68)

BRCA2 loss in the tumor contributes to the DNA repair defect (HRD). This is one of the key reasons PARP inhibitors are recommended for you.

Treatment Option: FDA Approved: olaparib, niraparib (PARP inhibitors) — HRD-positive

MYC

Pathogenic

Amplification (log2 = 1.85, chr8q24.21)

MYC is a gene that drives rapid cell growth. Amplification is found in about 30% of HGSOC tumors and is associated with aggressive disease. Investigational BET inhibitor drugs (like JQ1) target MYC-amplified cancers.

Treatment Option: BET inhibitors — preclinical evidence; not yet standard

CCNE1

Pathogenic

Amplification (log2 = 1.12, chr19q12)

CCNE1 amplification is a marker that the cancer cells are dividing very rapidly and may be less sensitive to platinum chemotherapy. CDK2 and Wee1 inhibitor drugs (such as adavosertib) target CCNE1-amplified cancers.

Treatment Option: CDK2 inhibitors, Wee1 inhibitors (adavosertib) — Phase I/II trials available

PIK3CA

Likely Pathogenic

Gain (log2 = 0.72, chr3)

PIK3CA gain activates the PI3K/AKT signaling pathway, which helps cancer cells survive. Combined with the PTEN deletion, this suggests the PI3K pathway is a treatment target.

Treatment Option: PI3K/AKT inhibitors — investigational in combination

CDKN2A

Pathogenic

Deletion (log2 = -1.42, chr9p21.3)

CDKN2A loss removes an important brake on the cell cycle. CDK4/6 inhibitors (like palbociclib) can help restore this brake and slow cancer cell division.

Treatment Option: CDK4/6 inhibitors (palbociclib) — off-label Phase II evidence; NCT06188520 recruiting

Your Tumor Environment

Advanced spatial analysis shows how different cells are organized in your tumor:

Tumor Microenvironment
Analysis of 300 spatial spots across four tissue regions (tumor core, stroma, immune zones, necrotic regions) reveals a mixed immune microenvironment with significant spatial exclusion of CD8+ T cells from the tumor core. Cancer-associated fibroblasts (CAFs) are highly enriched in the stroma with strong NNMT and FAP expression, suggesting a fibroblast barrier.

Immune Status: ● Mixed immune activity - Variable immune response across the tumor

Your tumor has immune cells (CD8+ T cells) that want to fight the cancer, but they are being blocked from entering the tumor by a protective barrier of fibroblast cells. This is why checkpoint inhibitor drugs alone may not be enough — targeting the fibroblast barrier (with NNMT inhibitors) alongside checkpoint therapy could help immune cells reach and destroy the cancer.

Key Patterns Observed

- CD8A expression: 923 TPM (immune zones) vs 154 TPM (tumor core) vs 29 TPM (stroma) — 31× exclusion gradient indicating immune cell spatial exclusion
- FAP = 755 TPM and NNMT = 453 TPM in stroma — CAF-high microenvironment confirmed
- Ki67 = 520 TPM and MYC = 512 TPM in tumor core — highly proliferative tumor
- PDL1 = 435 TPM in tumor core — checkpoint inhibitor target confirmed
- GZMB spatially clustered (Moran I = 0.018, Z = 4.41, $p<0.001$) — immune cytotoxic activity present but localized

Treatment Options

Based on your test results, your care team may consider these treatments:

Carboplatin + Paclitaxel

chemotherapy

NCCN Category 1

The standard combination chemotherapy for ovarian cancer at your stage. Carboplatin and paclitaxel work together to damage and kill cancer cells throughout the body.

Why this may help you: First-line treatment for Stage IIIC HGSOC per NCCN 2025 guidelines

Common side effects: Hair loss, Nausea, Fatigue, Numbness or tingling in hands and feet

Olaparib or Niraparib (PARP inhibitor maintenance)

targeted

FDA Approved

Expected Response: Significantly improved progression-free survival in HRD-positive HGSOC

PARP inhibitors block a backup DNA repair system that HRD-positive cancer cells rely on to survive. Your HRD score of 72 (above the 42 threshold) makes you a strong candidate for this maintenance therapy after chemotherapy.

Why this may help you: HRD score = 72 (positive; threshold ≥42); BRCA2 copy number loss; strongest actionable finding in genomic profile

Common side effects: Fatigue, Nausea, Anemia, Low platelet counts

Bevacizumab (anti-VEGF)

targeted

NCCN Category 1

Bevacizumab cuts off the blood supply that tumors need to grow. Your spatial tumor analysis shows elevated VEGFA expression in both the stroma (336 TPM) and tumor core (328 TPM), supporting its use.

Why this may help you: High-risk Stage IIIC features; elevated VEGFA spatially confirmed; NCCN-recommended for high-risk HGSOC

Common side effects: High blood pressure, Fatigue, Risk of wound healing problems

Neoantigen Vaccine (RMPEAAPPV — TP53 R175H peptide)

immunotherapy

Clinical Trial Only

A personalized vaccine designed to teach your immune system to recognize and attack cancer cells carrying a specific TP53 mutation. The peptide RMPEAAPPV was identified by AI analysis as a very strong match for your immune system (IC50 = 7.8 nM, strong binder).

Why this may help you: RMPEAAPPV: IC50 = 7.8 nM (strong HLA-A*02:01 binder); antigen presentation score = 0.85; TP53 loss confirmed; this is a novel finding not available through standard workup

Common side effects: Injection site reactions, Mild flu-like symptoms, Fatigue

NNMT Inhibitor (CAF-targeted)

targeted

Clinical Trial Only

A new class of drugs targeting cancer-associated fibroblasts (CAFs), which are protective cells in the tumor environment. Your spatial analysis shows very high NNMT (453 TPM) and FAP (755 TPM) in the stroma — indicating a CAF-rich barrier that may be blocking immune cells and other treatments.

Why this may help you: NNMT stroma = 453 TPM; FAP stroma = 755 TPM; CAF-high microenvironment confirmed; NNMT inhibition may improve drug and immune cell penetration into tumor

Common side effects: Under investigation — not yet well characterized

Anti-PD1/PDL1 Checkpoint Inhibitor + Combination

immunotherapy

Clinical Trial Only

Checkpoint inhibitors remove a molecular 'off switch' on immune cells, allowing them to attack the cancer. Your tumor shows high PDL1 (435 TPM in tumor core) but also significant CD8+ T cell exclusion — suggesting checkpoint inhibition combined with fibroblast targeting could allow immune cells to reach and kill cancer cells.

Why this may help you: PDL1 tumor core = 435 TPM; CD8A exclusion gradient = 31× (immune vs stroma); antigen presentation score = 0.85; spatial exclusion pattern identified as the key barrier to checkpoint therapy response

Common side effects: Immune-related side effects (rash, colitis, thyroid problems), Fatigue

Clinical Trials

You may be eligible for these research studies offering new treatments:

AZD8421 + CDK4/6 Inhibitors in HGSOC (AstraZeneca)
Phase: Phase I/II | **Status:** RECRUITING

Trial ID: NCT06188520

Your tumor shows CDK4 amplification and CDKN2A deletion — the exact molecular profile this trial is designed for. AstraZeneca is testing a new CDK4/6 inhibitor combination in HGSOC patients like you.

Nearest Location: 14 sites worldwide

SYN818 + Olaparib in HRD/BRCA Ovarian Cancer
Phase: Phase I | **Status:** RECRUITING

Trial ID: NCT07156253

Your HRD score of 72 and BRCA2 copy number loss make you potentially eligible for this combination trial testing a new drug alongside the PARP inhibitor olaparib.

Cirtuvivint + Olaparib in HRD Platinum-Resistant Ovarian Cancer
Phase: Phase I | **Status:** RECRUITING

Trial ID: NCT06856499

If your cancer becomes resistant to platinum chemotherapy, this trial tests a new combination targeting HRD-positive tumors like yours with olaparib plus a novel CLK/DYRK inhibitor.

VLS-1488 KIF18A Inhibitor in Chromosomal Instability Tumors
Phase: Phase I/II | **Status:** RECRUITING

Trial ID: NCT05902988

Your tumor has high chromosomal instability consistent with TP53 loss and HGSOC. This trial tests a novel drug that targets cancer cells with unstable chromosomes.

MATAO Phase III: Aromatase Inhibitor Maintenance in HGSOC
Phase: Phase III | **Status:** RECRUITING

Trial ID: NCT04111978

A large Phase III trial testing letrozole as maintenance therapy for HGSOC patients. With 53 recruiting sites, there is likely a location near you.

Nearest Location: 53 sites worldwide

Your Monitoring Plan

Regular check-ups help us track your health and catch any changes early.

Scheduled Tests

Every 3 months	CA-125 blood test Monitor for cancer recurrence or progression
Every 6 months or as indicated	CT scan (chest, abdomen, pelvis) Assess tumor response to treatment and detect recurrence
Before each chemotherapy cycle	Complete blood count (CBC) Monitor for anemia and low blood counts from treatment
Monthly during maintenance	PARP inhibitor toxicity labs Monitor kidney function and blood counts during PARP inhibitor therapy

⚠️ When to Call Your Doctor

Contact your care team right away if you experience:

- Sudden or worsening abdominal pain or bloating
- Unexplained weight loss or loss of appetite
- New shortness of breath or difficulty breathing
- Significantly worsening fatigue
- Fever above 38°C (100.4°F) while on chemotherapy

Questions? Your oncology nurse navigator or call the oncology triage line

For Your Family

BRCA2 copy number loss was detected in your tumor. While this appears to be a somatic (tumor-only) change, inherited BRCA2 mutations can affect first-degree relatives. Your children, siblings, and parents may benefit from genetic testing.

Recommended Steps for Family Members

- Consider referral to a certified genetic counselor for germline BRCA testing
- First-degree relatives (children, siblings, parents) may wish to discuss BRCA testing with their doctors

> **Genetic Counseling Recommended:** A genetic counselor can help explain what these results mean for your family members and discuss testing options.

Resources & Support

You don't have to face this alone. These resources can help:

📣 **Ovarian Cancer Research Alliance (OCRA)**
Leading ovarian cancer patient advocacy organization providing research updates, support groups, and clinical trial matching
| 🌐 https://ocrahope.org

👥 **Foundation for Women's Cancer**
Patient education, survivorship resources, and community support for gynecologic cancer patients
| 🌐 https://www.foundationforwomenscancer.org

📞 **National Cancer Institute Ovarian Cancer Resources**
Authoritative medical information about ovarian cancer, treatment options, and clinical trials from the NCI
| 🌐 https://www.cancer.gov/types/ovarian

⚠️ **Important Notice**

This AI-generated summary must be reviewed by your healthcare team before any treatment decisions. It is not a substitute for professional medical advice. All novel findings require prospective clinical validation.

Data Sources: mcp-mockepic, mcp-genomic-results, mcp-spatialtools, mcp-opentargets, mcp-neoantigen, mcp-clinicaltrials, mcp-geodownload
Report Version: 1.0
Generated: 2026-03-10 08:08:25

This AI-generated summary must be reviewed by your healthcare team before any treatment decisions. It is not a substitute for professional medical advice. All novel findings require prospective clinical validation.